

SOLIDWORKS API (.NET, VBA, C#) — Reference

(157/337): IMathUtility-IMathVector

SOLIDWORKS API Documentation (converted from CHM)

IMathUtility (Type)

IMathUtility Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMathUtility Interface

Provides access to the SOLIDWORKS math objects. These objects can simplify commonly used math calculations used with many API functions.
NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

Visual Basic Syntax

Visual Basic (Declaration)

```
Public Interface IMathUtility
```

Visual Basic (Usage)

```
Dim instance As IMathUtility
```

C#

```
public interface IMathUtility
```

C++/CLI

```
public interface class IMathUtility
```

Visual Basic for Applications (VBA) Syntax

See [MathUtility](#).

Example

[Align Assembly Component to Assembly Origin and Planes \(VBA\)](#)
[Create a Callout Independent of a Selection \(C#\)](#)
[Create a Callout Independent of a Selection \(VB.NET\)](#)
[Create a Callout Independent of a Selection \(VBA\)](#)

Accessors

[StdWorks::GetMathUtility](#)

Access Diagram

[MathUtility](#)

See Also

[IMathUtility Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMathPoint Interface](#)
[IMathTransform Interface](#)
[IMathVector Interface](#)
[IUserUnit Interface](#)

IMathUtility Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMathUtility Interface

The following tables list the members exposed by [IMathUtility](#).

Public Methods

Name	Description
ComposeTransform	Creates a new transform matrix.

		
	CreatePoint	Creates a new math point.
	CreateTransform	Creates a new math transform.
	CreateTransformRotateAxis	Creates a new math transform matrix that represents the rotation by an angle about a vector positioned at a point.
	CreateVector	Creates a math vector.
	ICreatePoint	Creates a new math point.
	ICreateTransform	Creates a new math transform.
	ICreateTransformRotateAxis	Creates a new math transform matrix that represents the rotation by an angle about a vector positioned at a point.
	ICreateVector	Creates a math vector.

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See Also

[IMathUtility Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMathPoint Interface](#)
[IMathTransform Interface](#)
[IMathVector Interface](#)
[IUserUnit Interface](#)

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IMathUtility Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMathUtility Interface

For a list of all members of this type, see [IMathUtility members](#).

Public Methods

Name	Description
	ComposeTransform Creates a new transform matrix.
	CreatePoint Creates a new math point.
	CreateTransform Creates a new math transform.
	CreateTransformRotateAxis Creates a new math transform matrix that represents the rotation by an angle about a vector positioned at a point.
	CreateVector Creates a math vector.
	ICreatePoint Creates a new math point.
	ICreateTransform Creates a new math transform.
	ICreateTransformRotateAxis Creates a new math transform matrix that represents the rotation by an angle about a vector positioned at a point.
	ICreateVector Creates a math vector.

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See Also

[IMathUtility Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMathPoint Interface](#)
[IMathTransform Interface](#)
[IMathVector Interface](#)
[IUserUnit Interface](#)

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ComposeTransform Method (IMathUtility)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathUtility Interface](#) : ComposeTransform Method (IMathUtility)

Creates a new transform matrix.

.NET Syntax

Visual Basic (Declaration)

```
Function ComposeTransform( _  
    ByVal XVec As MathVector, _  
    ByVal YVec As MathVector, _  
    ByVal ZVec As MathVector, _  
    ByVal TransVec As MathVector, _  
    ByVal Scale As System.Double _  
) As MathTransform
```

Visual Basic (Usage)



```
Dim instance As IMathUtility  
Dim XVec As MathVector  
Dim YVec As MathVector  
Dim ZVec As MathVector  
Dim TransVec As MathVector  
Dim Scale As System.Double  
Dim value As MathTransform
```

```
value = instance.ComposeTransform(XVec, YVec, ZVec, TransVec, Scale)
```

C#

```
MathTransform ComposeTransform(  
    MathVector XVec,  
    MathVector YVec,  
    MathVector ZVec,  
    MathVector TransVec,  
    System.double Scale  
)
```

C++/CLI

```
MathTransform^ ComposeTransform(  
& MathVector^ XVec,  
& MathVector^ YVec,  
& MathVector^ ZVec,  
& MathVector^ TransVec,  
& System.double Scale  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XVec

[X-axis of the transform origin](#)

YVec

[Y-axis of the transform origin](#)

ZVec

[Z-axis of the transform origin](#)

TransVec

Translation of the [origin](#) in space

Scale

Scale factor (typically 1; must be > 0)

Return Value

New [transform](#)

Visual Basic for Applications (VBA) Syntax

See [MathUtility::ComposeTransform](#).

Remarks

XVec, YVec and ZVec are expected to be mutually orthogonal. These three vectors can be non-unit vectors and are unitized during the construction of the [IMathTransform](#).

See Also

[IMathUtility Interface](#)
[IMathUtility Members](#)
[IMathUtility.CreateTransform Method](#)
[IMathUtility.CreateTransformRotateAxis Method](#)
[IMathUtility.ICreateTransform Method](#)
[IMathUtility.ICreateTransformRotateAxis Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreatePoint Method (IMathUtility)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathUtility Interface](#) : CreatePoint Method (IMathUtility)

Creates a new math point.

.NET Syntax

Visual Basic (Declaration)

```
Function CreatePoint( _  
    ByVal ArrayDataIn As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathUtility  
Dim ArrayDataIn As System.Object  
Dim value As System.Object  
  
value = instance.CreatePoint(ArrayDataIn)
```

C#

```
System.object CreatePoint(  
    System.object ArrayDataIn  
)
```

C++/CLI

```
System.Object^ CreatePoint(  
& System.Object^ ArrayDataIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArrayDataIn

Array of doubles representing the coordinates (x,y,z) of the point

Return Value

Newly created [IMathPoint](#) object or null if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathUtility.CreatePoint](#).

Example

[Dimension Edge in Drawing \(VBA\)](#)
[Get Names of Sketch Segments \(VBA\)](#)
[Transform Sketch to Model \(VBA\)](#)
[Zoom to Region \(VBA\)](#)
[Create a Callout Independent of a Selection \(VBA\)](#)
[Create a Callout Independent of a Selection \(C#\)](#)
[Create a Callout Independent of a Selection \(VB.NET\)](#)

See Also

[IMathUtility Interface](#)
[IMathUtility Members](#)
[IMathUtility.CreatePoint Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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CreateTransform Method (IMathUtility)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathUtility Interface](#) : CreateTransform Method (IMathUtility)

Creates a new math transform.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateTransform( _  
    ByVal ArrayDataIn As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathUtility  
Dim ArrayDataIn As System.Object  
Dim value As System.Object  
  
value = instance.CreateTransform(ArrayDataIn)
```

C#

```
System.object CreateTransform(  
    System.object ArrayDataIn  
)
```

C++/CLI

```
System.Object^ CreateTransform(  
& System.Object^ ArrayDataIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArrayDataIn

Sixteen (16) components of the transform (see **Remarks**)

Return Value

Newly created [math transform](#) or nothing or null if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathUtility.CreateTransform](#).

Example

[Align Assembly Component to Assembly Origin and Planes \(VBA\)](#)
[Use Presentation Transforms to Move Component \(VBA\)](#)
[Display Temporary Body \(C#\)](#)
[Display Temporary Body \(VB.NET\)](#)
[Display Temporary Body \(VBA\)](#)

Remarks

Transformation matrix data:

```
| a b c . n |  
| d e f . o |  
| g h i . p |  
| j k l . m |
```

The SOLIDWORKS transformation matrix is stored as a homogeneous matrix of 16 elements, ordered as shown. The first 9 elements (a to i) are elements of a 3x3 rotational sub-matrix, the next 3 elements (j, k, l) define a translation vector, and the next 1 element (m) is a scaling factor. The last 3 elements (n, o, p) are unused in this context.

The 3x3 rotational sub-matrix represents 3 axis sets:

- row 1 for x-axis components of rotation
- row 2 for y-axis components of rotation
- row 3 for z-axis components of rotation

The 3 axes are constrained to be orthogonal and unified so that they produce a pure rotational transformation. Reflections can also be added to these axes by setting the components to negative. The rotation sub-matrix coupled with the lower-left translation vector and the lower-right corner scaling factor creates an affine transformation, which is a transformation that preserves lines and parallelism; i.e., maps parallel lines to parallel lines.

If the 3 axis sets of the 3x3 rotational sub-matrix are not orthogonal or unified, then they are automatically corrected according to the following rules:

- If any axis is 0, or any two axes are parallel, or all axes are coplanar, then an identity matrix replaces the rotational sub-matrix.
- All axes are corrected to be of unit length.
- The axes are built to be orthogonal to each other in the prioritized order of Z, X, Y (X is orthogonal to Z, Y is orthogonal to Z and X).

See Also

[IMathUtility Interface](#)
[IMathUtility Members](#)
[IMathUtility::ICreateTransform Method](#)
[IMathUtility::ComposeTransform Method](#)
[IMathUtility::CreateTransformRotateAxis Method](#)
[IMathUtility::ICreateTransformRotateAxis Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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CreateTransformRotateAxis Method (IMathUtility)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathUtility Interface](#) : CreateTransformRotateAxis Method (IMathUtility)

Creates a new math transform matrix that represents the rotation by an angle about a vector positioned at a point.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateTransformRotateAxis( _  
    ByVal PointObjIn As System.Object, _  
    ByVal VectorObjIn As System.Object, _  
    ByVal Angle As System.Double _  
    ) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathUtility  
Dim PointObjIn As System.Object  
Dim VectorObjIn As System.Object  
Dim Angle As System.Double  
Dim value As System.Object  
  
value = instance.CreateTransformRotateAxis(PointObjIn, VectorObjIn, Angle)
```

C#

```
System.Object CreateTransformRotateAxis(  
    System.Object PointObjIn,  
    System.Object VectorObjIn,  
    System.Double Angle  
)
```

C++/CLI

```
System.Object^ CreateTransformRotateAxis(  
& System.Object^ PointObjIn,  
& System.Object^ VectorObjIn,  
& System.Double Angle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointObjIn

Center [point](#) of the axis of the transform

VectorObjIn

Axis [vector](#) of the transform

Angle

Angle of rotation about the X axis vector

Return Value

Newly created [math transform](#) or null or nothing if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathUtility::CreateTransformRotateAxis](#).

Example

[Rotate Assembly Component on Axis Using IDragOperator::Drag \(VBA\)](#)
[Rotate Assembly Component on Axis Using IDragOperator::Drag \(VB.NET\)](#)
[Rotate Assembly Component on Axis Using IDragOperator::Drag \(C#\)](#)
[Rotate Assembly Component on Axis Using IDragOperator::DragAsUI \(VBA\)](#)
[Rotate Assembly Component on Axis Using IDragOperator::DragAsUI \(VB.NET\)](#)
[Rotate Assembly Component on Axis Using IDragOperator::DragAsUI \(C#\)](#)

See Also

[IMathUtility Interface](#)
[IMathUtility Members](#)
[IMathUtility::Compose Transform Method](#)
[IMathUtility::Create Transform Method](#)
[IMathUtility::ICreate Transform Method](#)
[IMathUtility::ICreate Transform Rotate Axis Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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CreateVector Method (IMathUtility)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathUtility Interface](#) : CreateVector Method (IMathUtility)

Creates a math vector.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateVector( _  
    ByVal ArrayDataIn As System.Object _  
) As System.Object
```

Visual Basic (Usage)

 Copy Code

```
Dim instance As IMathUtility
Dim ArrayDataIn As System.Object
Dim value As System.Object

value = instance.CreateVector(ArrayDataIn)
```

C#

```
System.object CreateVector(
    System.object ArrayDataIn
)
```

C++/CLI

```
System.Object^ CreateVector(
    & System.Object^ ArrayDataIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArrayDataIn

Array of doubles representing the x,y,z components of the vector

Return Value

Newly created [math vector](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathUtility::CreateVector](#).

Example

[Create Temporary Extruded Body \(VBA\)](#)
[Get Angle of Hole Not Normal to a Face \(VBA\)](#)
[Locate Apex of Conical Face \(VBA\)](#)
[Create Drag Arrow Manipulator \(C#\)](#)
[Create Drag Arrow Manipulator \(VB.NET\)](#)
[Create Drag Arrow Manipulator \(VBA\)](#)

See Also

[IMathUtility Interface](#)
[IMathUtility Members](#)
[IMathUtility::!CreateVector Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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ICreatePoint Method (IMathUtility)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathUtility Interface](#) : ICreatePoint Method (IMathUtility)

Creates a new math point.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreatePoint( _
    ByRef ArrayDataIn As System.Double _
) As MathPoint
```

Visual Basic (Usage)



```
Dim instance As IMathUtility
Dim ArrayDataIn As System.Double
Dim value As MathPoint

value = instance.ICreatePoint(ArrayDataIn)
```

C#

```
MathPoint ICreatePoint(
    ref System.double ArrayDataIn
)
```

C++/CLI

```
MathPoint^ ICreatePoint(
    & System.double% ArrayDataIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArrayDataIn

Array of doubles representing the coordinates (x,y,z) of the point

Return Value

Newly created [IMathPoint](#) object or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathUtility::ICreatePoint](#).

See Also

[IMathUtility Interface](#)
[IMathUtility Members](#)
[IMathUtility::CreatePoint Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateTransform Method (IMathUtility)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathUtility Interface](#) : ICreateTransform Method (IMathUtility)

Creates a new math transform.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateTransform( _
    ByRef ArrayDataIn As System.Double _
) As MathTransform
```

Visual Basic (Usage)

```
Dim instance As IMathUtility
Dim ArrayDataIn As System.Double
Dim value As MathTransform

value = instance.ICreateTransform(ArrayDataIn)
```

C#



```
MathTransform ICreateTransform(
    ref System.double ArrayDataIn
)
```

C++/CLI

```
MathTransform^ ICreateTransform(
    & System.double% ArrayDataIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArrayDataIn

- in-process, unmanaged C++: Pointer to sixteen (16) components of the transform (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Return Value

Newly created [math transform](#) or null if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathUtility::ICreateTransform](#).

Remarks

Transformation matrix data:

```
| a b c . n |
| d e f . o |
| g h i . p |
| j k l . m |
```

The SOLIDWORKS transformation matrix is stored as a homogeneous matrix of 16 elements, ordered as shown. The first 9 elements (a to i) are elements of a 3x3 rotational sub-matrix, the next 3 elements (j, k, l) define a translation vector, and the next 1 element (m) is a scaling factor. The last 3 elements (n, o, p) are unused in this context.

The 3x3 rotational sub-matrix represents 3 axis sets:

- row 1 for x-axis components of rotation
- row 2 for y-axis components of rotation
- row 3 for z-axis components of rotation

The 3 axes are constrained to be orthogonal and unified so that they produce a pure rotational transformation. Reflections can also be added to these axes by setting the components to negative. The rotation sub-matrix coupled with the lower-left translation vector and the lower-right corner scaling factor creates an affine transformation, which is a transformation that preserves lines and parallelism; i.e., maps parallel lines to parallel lines.

If the 3 axis sets of the 3x3 rotational sub-matrix are not orthogonal or unified, then they are automatically corrected according to the following rules:

- If any axis is 0, or any two axes are parallel, or all axes are coplanar, then an identity matrix replaces the rotational sub-matrix.
- All axes are corrected to be of unit length.
- The axes are built to be orthogonal to each other in the prioritized order of Z, X, Y (X is orthogonal to Z, Y is orthogonal to Z and X).

See Also

[IMathUtility Interface](#)
[IMathUtility Members](#)
[IMathUtility::CreatePoint Method](#)
[IMathUtility::ComposeTransform Method](#)
[IMathUtility::CreateTransformRotateAxis Method](#)
[IMathUtility::ICreateTransformRotateAxis Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathUtility Interface](#) : ICreateTransformRotateAxis Method (IMathUtility)

Creates a new math transform matrix that represents the rotation by an angle about a vector positioned at a point.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateTransformRotateAxis( _  
    ByVal PointObjIn As MathPoint, _  
    ByVal VectorObjIn As MathVector, _  
    ByVal Angle As System.Double _  
) As MathTransform
```

Visual Basic (Usage)



```
Dim instance As IMathUtility  
Dim PointObjIn As MathPoint  
Dim VectorObjIn As MathVector  
Dim Angle As System.Double  
Dim value As MathTransform  
  
value = instance.ICreateTransformRotateAxis(PointObjIn, VectorObjIn, Angle)
```

C#

```
MathTransform ICreateTransformRotateAxis(  
    MathPoint PointObjIn,  
    MathVector VectorObjIn,  
    System.double Angle  
)
```

C++/CLI

```
MathTransform^ ICreateTransformRotateAxis(  
& MathPoint^ PointObjIn,  
& MathVector^ VectorObjIn,  
& System.double Angle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointObjIn

Center [point](#) of the axis of the transform

VectorObjIn

Axis [vector](#) of the transform

Angle

Angle of rotation about the X axis vector

Return Value

Newly created [math transform](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathUtility::ICreateTransformRotateAxis](#).

See Also

[IMathUtility Interface](#)

[IMathUtility Members](#)

[IMathUtility::CreateTransformRotateAxis Method](#)

[IMathUtility::ComposeTransform Method](#)

[IMathUtility::CreateTransform Method](#)

[IMathUtility::!CreateTransform Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

ICreateVector Method (IMathUtility)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathUtility Interface](#) : ICreateVector Method (IMathUtility)

Creates a math vector.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateVector( _  
    ByRef ArrayDataIn As System.Double _  
) As MathVector
```

Visual Basic (Usage)



```
Dim instance As IMathUtility  
Dim ArrayDataIn As System.Double  
Dim value As MathVector  
  
value = instance.ICreateVector(ArrayDataIn)
```

C#

```
MathVector ICreateVector(  
    ref System.double ArrayDataIn  
)
```

C++/CLI

```
MathVector^ ICreateVector(  
& System.double% ArrayDataIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArrayDataIn

- in-process, unmanaged C++: Pointer to an array of doubles representing the x,y,z components of the vector

• VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

Newly created [math vector](#) or NULL if the operation fails

See Also

[IMathUtility Interface](#)
[IMathUtility Members](#)
[IMathUtility::CreateVector Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

IMathVector (Type)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IMathVector Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMathVector Interface

Provides a simplified interface for manipulating math vectors.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IMathVector
```

Visual Basic (Usage)



```
Dim instance As IMathVector
```

C#

```
public interface IMathVector
```

C++/CLI

```
public interface class IMathVector
```

Visual Basic for Applications (VBA) Syntax

See [MathVector](#).

Example

[Get Angle Between Plane and Line \(VBA\)](#)
[Get Body Outline \(C#\)](#)
[Get Body Outline \(VB.NET\)](#)
[Get Body Outline \(VBA\)](#)

Accessors

[ICamera::GetUpVector](#)
[ICamera::GetViewVector](#)
[IDimension::DimensionLineDirection](#)
[IDimension::ExtensionLineDirection](#)
[IDragArrowManipulator::Direction](#)
[IMathPoint::ConvertToVector](#) and [IMathPoint::!ConvertToVector](#)
[IMathPoint::Subtract](#) and [IMathPoint::!Subtract](#)
[IMathTransform::GetData2](#) and [IMathTransform::!GetData2](#)
[IMathUtility::CreateVector](#) and [IMathUtility::!CreateVector](#)
[IMathVector::Add](#) and [IMathVector::!Add](#)
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[IMathVector::Normalise](#)
[IMathVector::Scale](#) and [IMathVector::!Scale](#)
[IMathVector::Subtract](#) and [IMathVector::!Subtract](#)
[IModelView::Translation3](#)
[IPlaneManipulator::Normal](#)
[IRuledSurfaceFeatureData::DirectionVector](#)

- [ISectionViewData::GetFirstPlaneParameters](#)
- [ISectionViewData::GetSecondPlaneParameters](#)
- [ISectionViewData::GetThirdPlaneParameters](#)
- [ISwScene::GetFloorNormal](#)
- [ITriadManipulator::XAxis](#)
- [ITriadManipulator::YAxis](#)
- [ITriadManipulator::ZAxis](#)
- [IView3D::Translation](#)

 Access Diagram

[MathVector](#)

 See Also

- [IMathVector Members](#)
- [SolidWorks.Interop.sldworks Namespace](#)
- [IMathPoint Interface](#)
- [IMathTransform Interface](#)
- [IMathUtility Interface](#)
- [IUserUnit Interface](#)

IMathVector Interface Members

[See Also](#) [Properties](#) [Methods](#)

 [SolidWorks.Interop.sldworks Namespace](#) : IMathVector Interface

The following tables list the members exposed by [IMathVector](#).

 Public Properties

Name	Description
 ArrayData	Gets or sets the array of three doubles for this vector.
 IArrayData	Gets or sets the array of three doubles for this vector.

[Top](#)

 Public Methods

Name	Description
 Add	Adds this math vector with the specified math vector.
 ConvertToPoint	Converts this math vector to a point by using the three components of the math vector to be the coordinates of a new math point.
 Cross	Gets the cross product between two math vectors.
 Dot	Gets the value of the dot product between two math vectors.
 GetLength	Gets the length of a math vector.
 IAdd	Adds this math vector with the specified math vector.
 IConvertToPoint	Converts this math vector to a point by using the three components of the math vector to be the coordinates of a new math point.
 ICross	Gets the cross product between two math vectors.
 IDot	Gets the value of the dot product between two math vectors.
 IMultiplyTransform	Multiplies this math vector by the specified math transform.
 IScale	Scales this math vector's magnitude by a factor and returns a new math vector.
 ISubtract	Subtracts the specified math vector's magnitude from this math vector and returns a new math vector.
 MultiplyTransform	Multiplies this math vector by the specified math transform.
 Normalise	Gets the unit-length vector for this math vector.
 Scale	Scales this math vector's magnitude by a factor and returns a new math vector.
 Subtract	Subtracts the specified math vector's magnitude from this math vector and returns a new math vector.



[Top](#)

See Also

[IMathVector Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMathPoint Interface](#)
[IMathTransform Interface](#)
[IMathUtility Interface](#)
[IUserUnit Interface](#)

SOLIDWORKS API Help

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IMathVector Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMathVector Interface

For a list of all members of this type, see [IMathVector members](#).

Public Methods

Name	Description
Add	Adds this math vector with the specified math vector.
ConvertToPoint	Converts this math vector to a point by using the three components of the math vector to be the coordinates of a new math point.
Cross	Gets the cross product between two math vectors.
Dot	Gets the value of the dot product between two math vectors.
GetLength	Gets the length of a math vector.
IAdd	Adds this math vector with the specified math vector.
IConvertToPoint	Converts this math vector to a point by using the three components of the math vector to be the coordinates of a new math point.
ICross	Gets the cross product between two math vectors.
IDot	Gets the value of the dot product between two math vectors.
IMultiplyTransform	Multiplies this math vector by the specified math transform.
IScale	Scales this math vector's magnitude by a factor and returns a new math vector.
ISubtract	Subtracts the specified math vector's magnitude from this math vector and returns a new math vector.
MultiplyTransform	Multiplies this math vector by the specified math transform.
Normalise	Gets the unit-length vector for this math vector.
Scale	Scales this math vector's magnitude by a factor and returns a new math vector.
Subtract	Subtracts the specified math vector's magnitude from this math vector and returns a new math vector.

[Top](#)

See Also

[IMathVector Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMathPoint Interface](#)
[IMathTransform Interface](#)
[IMathUtility Interface](#)
[IUserUnit Interface](#)

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IMathVector Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMathVector Interface

For a list of all members of this type, see [IMathVector members](#).

Public Properties

Name	Description
------	-------------



[ArrayData](#)

Gets or sets the array of three doubles for this vector.



[IArrayData](#)

Gets or sets the array of three doubles for this vector.

[Top](#)

See Also

[IMathVector Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMathPoint Interface](#)
[IMathTransform Interface](#)
[IMathUtility Interface](#)
[IUserUnit Interface](#)

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Add Method (IMathVector)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : Add Method (IMathVector)

Adds this math vector with the specified math vector.

.NET Syntax

Visual Basic (Declaration)

```
Function Add( _  
    ByVal VectorObjIn As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim VectorObjIn As System.Object  
Dim value As System.Object  
  
value = instance.Add(VectorObjIn)
```

C#

```
System.object Add(  
    System.object VectorObjIn  
)
```

C++/CLI

```
System.Object^ Add(  
& System.Object^ VectorObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

VectorObjIn

[Math vector](#) object to add to this math vector

Return Value

Newly created [math vector](#) object or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathVector::Add](#).

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[IMathVector::IAdd Method](#)
[IMathVector::Subtract Method](#)
[IMathVector::ISubtract Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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ArrayData Property (IMathVector)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : ArrayData Property (IMathVector)

Gets or sets the array of three doubles for this vector.

.NET Syntax

Visual Basic (Declaration)

Property ArrayData As System.Object

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim value As System.Object
```

```
instance.ArrayData = value
```

```
value = instance.ArrayData
```

C#

```
System.object ArrayData {get; set;}
```

C++/CLI

```
property System.Object^ ArrayData {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of three doubles

Visual Basic for Applications (VBA) Syntax

See [MathVector::ArrayData](#).

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[IMathVector::IArrayData Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ConvertToPoint Method (IMathVector)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : ConvertToPoint Method (IMathVector)

Converts this math vector to a point by using the three components of the math vector to be the coordinates of a new math point.

.NET Syntax

Visual Basic (Declaration)

Function ConvertToPoint() As System.Object

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim value As System.Object  
  
value = instance.ConvertToPoint()
```

C#

```
System.object ConvertToPoint()
```

C++/CLI

```
System.Object^ ConvertToPoint();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Newly created [IMathPoint](#) object or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathVector::ConvertToPoint](#).

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[IMathVector::ConvertToPoint Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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Cross Method (IMathVector)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : Cross Method (IMathVector)

Gets the cross product between two math vectors.

.NET Syntax

Visual Basic (Declaration)

```
Function Cross(  
    ByVal VectorObjIn As System.Object  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim VectorObjIn As System.Object  
Dim value As System.Object  
  
value = instance.Cross(VectorObjIn)
```

C#

```
System.object Cross(  
    System.object VectorObjIn  
)
```

C++/CLI

```
System.Object^ Cross(  
& System.Object^ VectorObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

VectorObjIn
[Math vector](#) to use to determine the cross product

Return Value

Newly created [math vector](#)

Visual Basic for Applications (VBA) Syntax

See [MathVector::Cross](#).

Example

[Get Angle Between Plane and Line \(VBA\)](#)

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[IMathVector::ICross Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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Dot Method (IMathVector)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : Dot Method (IMathVector)

Gets the value of the dot product between two math vectors.

.NET Syntax

Visual Basic (Declaration)

```
Function Dot( _  
    ByVal VectorObjIn As System.Object _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim VectorObjIn As System.Object  
Dim value As System.Double  
  
value = instance.Dot(VectorObjIn)
```

C#

```
System.double Dot(  
    System.object VectorObjIn  
)
```

C++/CLI

```
System.double Dot(  
& System.Object^ VectorObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

VectorObjIn

[Math vector](#) to use to determine the dot value

Return Value

Value of the dot product

Visual Basic for Applications (VBA) Syntax

See [MathVector::Dot](#).

Example

[Get Angle of Hole Not Normal to Face \(VBA\)](#)

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[IMathVector::IDot Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

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GetLength Method (IMathVector)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : GetLength Method (IMathVector)

Gets the length of a math vector.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLength() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim value As System.Double  
  
value = instance.GetLength()
```

C#

```
System.double GetLength()
```

C++/CLI

```
System.double GetLength();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Length of [math vector](#)

Visual Basic for Applications (VBA) Syntax

See [MathVector::GetLength](#).

Example

[Get Angle Between Plane and Line \(VBA\)](#)

See Also

[IMathVector Interface](#)
[IMathVector Members](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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IAdd Method (IMathVector)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : IAdd Method (IMathVector)

Adds this math vector with the specified math vector.

.NET Syntax

Visual Basic (Declaration)

```
Function IAdd( _  
    ByVal VectorObjIn As MathVector _  
) As MathVector
```

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim VectorObjIn As MathVector  
Dim value As MathVector  
  
value = instance.IAdd(VectorObjIn)
```

C#

```
MathVector IAdd(  
    MathVector VectorObjIn  
)
```

C++/CLI

```
MathVector^ IAdd(  
& MathVector^ VectorObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

VectorObjIn

[Math vector](#) object to add to this math vector

Return Value

Newly created [math vector](#) object or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathVector::IAdd](#).

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[MathVector::Add Method](#)
[IMathVector::ISubtract Method](#)
[IMathVector::Subtract Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

IArrayData Property (IMathVector)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : IArrayData Property (IMathVector)

Gets or sets the array of three doubles for this vector.

.NET Syntax

Visual Basic (Declaration)

Property IArrayData As System.Double

Visual Basic (Usage)



```
Dim instance As IMathVector
Dim value As System.Double

instance.IArrayData = value

value = instance.IArrayData
```

C#

```
System.double IArrayData {get; set;}
```

C++/CLI

```
property System.double IArrayData {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

- in-process, unmanaged C++: Pointer to an array of three doubles
- VBA, VB.NET, C#, and C++/CLI: Not supported

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[IMathVector.IArrayData Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

IConvertToPoint Method (IMathVector)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : IConvertToPoint Method (IMathVector)

Converts this math vector to a point by using the three components of the math vector to be the coordinates of a new math point.

.NET Syntax

Visual Basic (Declaration)

Function IConvertToPoint() As MathPoint

Visual Basic (Usage)



```
Dim instance As IMathVector
Dim value As MathPoint
```

```
value = instance.IConvertToPoint()
```

C#

```
MathPoint IConvertToPoint()
```

C++/CLI

```
MathPoint^ IConvertToPoint();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Newly created [IMathPoint](#) object or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathVector::IConvertToPoint](#).

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[IMathVector::ConvertToPoint Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

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ICross Method (IMathVector)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : ICross Method (IMathVector)

Gets the cross product between two math vectors.

.NET Syntax

Visual Basic (Declaration)

```
Function ICross(  
    ByVal VectorObjIn As MathVector  
) As MathVector
```

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim VectorObjIn As MathVector  
Dim value As MathVector
```

```
value = instance.ICross(VectorObjIn)
```

C#

```
MathVector ICross(  
    MathVector VectorObjIn  
)
```

C++/CLI

```
MathVector^ ICross(  
& MathVector^ VectorObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

VectorObjIn

[Math vector](#) to use to determine the cross product

Return Value

Newly created [math vector](#)

Visual Basic for Applications (VBA) Syntax

See [MathVector::ICross](#).

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[IMathVector::Cross Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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IDot Method (IMathVector)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : IDot Method (IMathVector)

Gets the value of the dot product between two math vectors.

.NET Syntax

Visual Basic (Declaration)

```
Function IDot( _  
    ByVal VectorObjIn As MathVector _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim VectorObjIn As MathVector  
Dim value As System.Double  
  
value = instance.IDot(VectorObjIn)
```

C#

```
System.double IDot(  
    MathVector VectorObjIn  
)
```

C++/CLI

```
System.double IDot(  
    & MathVector^ VectorObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

VectorObjIn
[Math vector](#) to use to determine the dot value

Return Value

Value of the dot product

Visual Basic for Applications (VBA) Syntax

See [MathVector::IDot](#).

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[IMathVector::Dot Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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MultiplyTransform Method (IMathVector)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : IMultiplyTransform Method (IMathVector)

Multiplies this math vector by the specified math transform.

.NET Syntax

Visual Basic (Declaration)

```
Function IMultiplyTransform( _  
    ByVal TransformObjIn As MathTransform _  
) As MathVector
```

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim TransformObjIn As MathTransform  
Dim value As MathVector  
  
value = instance.IMultiplyTransform(TransformObjIn)
```

C#

```
MathVector IMultiplyTransform(  
    MathTransform TransformObjIn  
)
```

C++/CLI

```
MathVector^ IMultiplyTransform(  
& MathTransform^ TransformObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TransformObjIn

[Math transform](#) by which to multiply the math vector

Return Value

Newly created [math vector](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathVector::IMultiplyTransform](#).

Remarks

The matrix is rotated and then scaled. It is not transformed because the math vector is assumed to be positionless.

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[IMathVector::MultiplyTransform Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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IScale Method (IMathVector)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : IScale Method (IMathVector)

Scales this math vector's magnitude by a factor and returns a new math vector.

.NET Syntax

Visual Basic (Declaration)

```
Function IScale( _  
    ByVal ValueIn As System.Double _  
) As MathVector
```

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim ValueIn As System.Double  
Dim value As MathVector  
  
value = instance.IScale(ValueIn)
```

C#

```
MathVector IScale(  
    System.double ValueIn  
)
```

C++/CLI

```
MathVector^ IScale(  
& System.double ValueIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ValueIn

Value by which to scale the math vector

Return Value

Newly created [math vector](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathVector::IScale](#).

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[IMathVector::Scale Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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ISubtract Method (IMathVector)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : ISubtract Method (IMathVector)

Subtracts the specified math vector's magnitude from this math vector and returns a new math vector.

.NET Syntax

Visual Basic (Declaration)

```
Function ISubtract( _  
    ByVal VectorObjIn As MathVector _  
) As MathVector
```

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim VectorObjIn As MathVector  
Dim value As MathVector  
  
value = instance.ISubtract(VectorObjIn)
```

C#

```
MathVector ISubtract(  
    MathVector VectorObjIn  
)
```

C++/CLI

```
MathVector^ ISubtract(  
& MathVector^ VectorObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

VectorObjIn

[Math vector](#) to subtract from this math vector

Return Value

Newly created [math vector](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathVector::ISubtract](#).

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[IMathVector::Subtract Method](#)
[IMathVector::Add Method](#)
[IMathVector::IAdd Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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MultiplyTransform Method (IMathVector)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : MultiplyTransform Method (IMathVector)

Multiplies this math vector by the specified math transform.

.NET Syntax

Visual Basic (Declaration)

```
Function MultiplyTransform( _  
    ByVal TransformObjIn As System.Object _
```

```
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim TransformObjIn As System.Object  
Dim value As System.Object  
  
value = instance.MultiplyTransform(TransformObjIn)
```

C#

```
System.object MultiplyTransform(  
    System.object TransformObjIn  
)
```

C++/CLI

```
System.Object^ MultiplyTransform(  
& System.Object^ TransformObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TransformObjIn

[Math transform](#) by which to multiply the math vector

Return Value

Newly created [math vector](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathVector::MultiplyTransform](#).

Example

[Get Angle Between Plane and Line \(VBA\)](#)

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[IMathVector::IMultiplyTransform Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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Normalise Method (IMathVector)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : Normalise Method (IMathVector)

Gets the unit-length vector for this math vector.

.NET Syntax

Visual Basic (Declaration)

```
Function Normalise() As MathVector
```

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim value As MathVector
```

```
value = instance.Normalise()
```

C#

```
MathVector Normalise()
```

C++/CLI

```
MathVector^ Normalise();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Unit-length vector

Visual Basic for Applications (VBA) Syntax

See [MathVector::Normalise](#).

See Also

[IMathVector Interface](#)
[IMathVector Members](#)

Availability

SOLIDWORKS 2001Plus SP2, Revision Number 10.2

SOLIDWORKS API Help

[Send comments](#) on this topic.

Scale Method (IMathVector)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : Scale Method (IMathVector)

Scales this math vector's magnitude by a factor and returns a new math vector.

.NET Syntax

Visual Basic (Declaration)

```
Function Scale( _  
    ByVal ValueIn As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim ValueIn As System.Double  
Dim value As System.Object  
  
value = instance.Scale(ValueIn)
```

C#

```
System.object Scale(  
    System.double ValueIn  
)
```

C++/CLI

```
System.Object^ Scale(  
&    System.double ValueIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ValueIn

Value by which to scale the math vector

Return Value

Newly created [math vector](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathVector::Scale](#).

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[IMathVector::IScale Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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Subtract Method (IMathVector)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathVector Interface](#) : Subtract Method (IMathVector)

Subtracts the specified math vector's magnitude from this math vector and returns a new math vector.

.NET Syntax

Visual Basic (Declaration)

```
Function Subtract( _  
    ByVal VectorObjIn As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathVector  
Dim VectorObjIn As System.Object  
Dim value As System.Object
```

```
value = instance.Subtract(VectorObjIn)
```

C#

```
System.object Subtract(  
    System.object VectorObjIn  
)
```

C++/CLI

```
System.Object^ Subtract(  
    & System.Object^ VectorObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

VectorObjIn

[Math vector](#) to subtract from this math vector

Return Value

Newly created [math vector](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathVector::Subtract](#).

See Also

[IMathVector Interface](#)
[IMathVector Members](#)
[IMathVector::ISubtract Method](#)
[IMathVector::Add Method](#)
[IMathVector::IAdd Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0